

Some properties of the logical connectives in PL

Let us choose and fix any tautology in the set $\mathcal{F}$ of all wffs of PL (over some set PV of propositional variables) and denote it by $\top$.

So for every tautology α , we have $\alpha \equiv \top$.

Similarly, let us fix the notation $\perp$ for a contradiction.

Let us recall the following properties of the logical connectives $\wedge, \vee, \neg$ that we have mentioned earlier.

1. $\alpha \wedge \beta \equiv \beta \wedge \alpha$
2. $\alpha \vee \beta \equiv \beta \vee \alpha$
3. $\alpha \wedge (\beta \wedge \gamma) \equiv (\alpha \wedge \beta) \wedge \gamma$
4. $\alpha \vee (\beta \vee \gamma) \equiv (\alpha \vee \beta) \vee \gamma$
5. $\alpha \wedge (\beta \vee \gamma) \equiv (\alpha \wedge \beta) \vee (\alpha \wedge \gamma)$
6. $\alpha \vee (\beta \wedge \gamma) \equiv (\alpha \vee \beta) \wedge (\alpha \vee \gamma)$
7. $\neg\neg\alpha \equiv \alpha$
8. $\neg(\alpha \wedge \beta) \equiv \neg\alpha \vee \neg\beta$
9. $\neg(\alpha \vee \beta) \equiv \neg\alpha \wedge \neg\beta$.
10. $\alpha \wedge \top \equiv \alpha$
11. $\alpha \vee \top \equiv \top$
12. $\alpha \wedge \perp \equiv \perp$
13. $\alpha \vee \perp \equiv \alpha$
14. $\neg\top \equiv \perp$
15. $\neg\perp \equiv \top$

These seem to indicate some familiar properties: 1-4 appear to refer to commutativity and associativity of the ‘operators’ $\wedge, \vee$ in the set $\mathcal{F}$, 5-6 indicate distributivity of $\wedge, \vee$ over each other, 7 indicates that $\neg$ has the property of *involution*, 8-9 say that $\wedge, \vee, \neg$ satisfy the DeMorgan laws.

The only catch is that the above are not *equations* in $\mathcal{F}$, but *logical equivalences* between wffs in $\mathcal{F}$:
for $\alpha \neq \beta$, $\alpha \wedge \beta \neq \beta \wedge \alpha$

Can we turn all the above logical equivalences into equations in some set?

1 Operators in the quotient set $\mathcal{F}/\equiv$

Define the following operators in $\mathcal{F}/\equiv$.
Let $[\alpha], [\beta]$ be any elements in $\mathcal{F}/\equiv$.

Definition 1.1.

- (i) $[\alpha] \wedge [\beta] := [\alpha \wedge \beta]$.
- (ii) $[\alpha] \vee [\beta] := [\alpha \vee \beta]$.
- (iii) $\neg[\alpha] := [\neg\alpha]$.

Note. We have abused notation, using $\wedge, \vee, \neg$ for the logical connectives as well as the operators in $\mathcal{F}/\equiv$. The context should make the usage clear.

Clearly, we need to ascertain that (i)-(iii) are well-defined.
In other words, we need to prove that

Exercise 1.1.

- (i) $[\alpha] \wedge [\beta] = [\alpha'] \wedge [\beta']$,
 - (ii) $[\alpha] \vee [\beta] = [\alpha'] \vee [\beta']$,
 - (iii) $\neg[\alpha] = \neg[\alpha']$,
- for any $\alpha' \in [\alpha], \beta' \in [\beta]$ (i.e. $\alpha \equiv \alpha', \beta \equiv \beta'$).

We can now show that, all the logical equivalences listed at the beginning of this section, translate into equations in the set $\mathcal{F}/\equiv$. For instance,

$$\begin{aligned} [\alpha] \wedge [\beta] &= [\beta] \wedge [\alpha], \\ [\alpha] \wedge [\top] &= [\alpha], \text{ or} \\ \neg([\alpha] \wedge [\beta]) &= \neg[\alpha] \vee \neg[\beta]. \end{aligned}$$

(Exercise.)

Moreover, as noted earlier,

- (i) the equivalence class $[\top]$ is the set of exactly the tautologies, and
- (ii) $[\perp]$ is the set of exactly the contradictions.

Notation. We reserve the notation ‘1’ for $[\top]$, and ‘0’ for $[\perp]$. The reason will be clear in the sequel.

2 The equivalence relation $\sim$ on $\mathcal{F}$

Recall the equivalence relation $\sim$ on $\mathcal{F}$ defined using the syntactical version of logical equivalence, viz.

$$\alpha \sim \beta \text{ if and only if } \vdash \alpha \leftrightarrow \beta,$$

for any wffs α, β in $\mathcal{F}$.

Observe that operators $\wedge, \vee, \neg$ can be defined identically as in Definition ??, on the quotient set $\mathcal{F}/\sim$. That the operations are well-defined, has to be proved here with the help of the properties of $\vdash$. Note that, here, the *theorems* of PL constitute an equivalence class in $\mathcal{F}/\sim$. As observed earlier, the quotient sets $\mathcal{F}/\equiv$ and $\mathcal{F}/\sim$ are identical, but nevertheless, $\sim$ allows us to take a syntactic look at the equivalence classes of wffs, as opposed to the semantic look imparted by $\equiv$.

The above deliberations on $\mathcal{F}/\equiv$ (or $\mathcal{F}/\sim$) appear to suggest an algebraic structure for the set. Let us investigate this now. We shall find that $\mathcal{F}/\equiv$ forms what is called a *Boolean algebra*, and we will finally be led to an close connection between the class of all Boolean algebras, and PL.